Designing Augmented Futures: AI, Accessibility, and Critical Making in TCID

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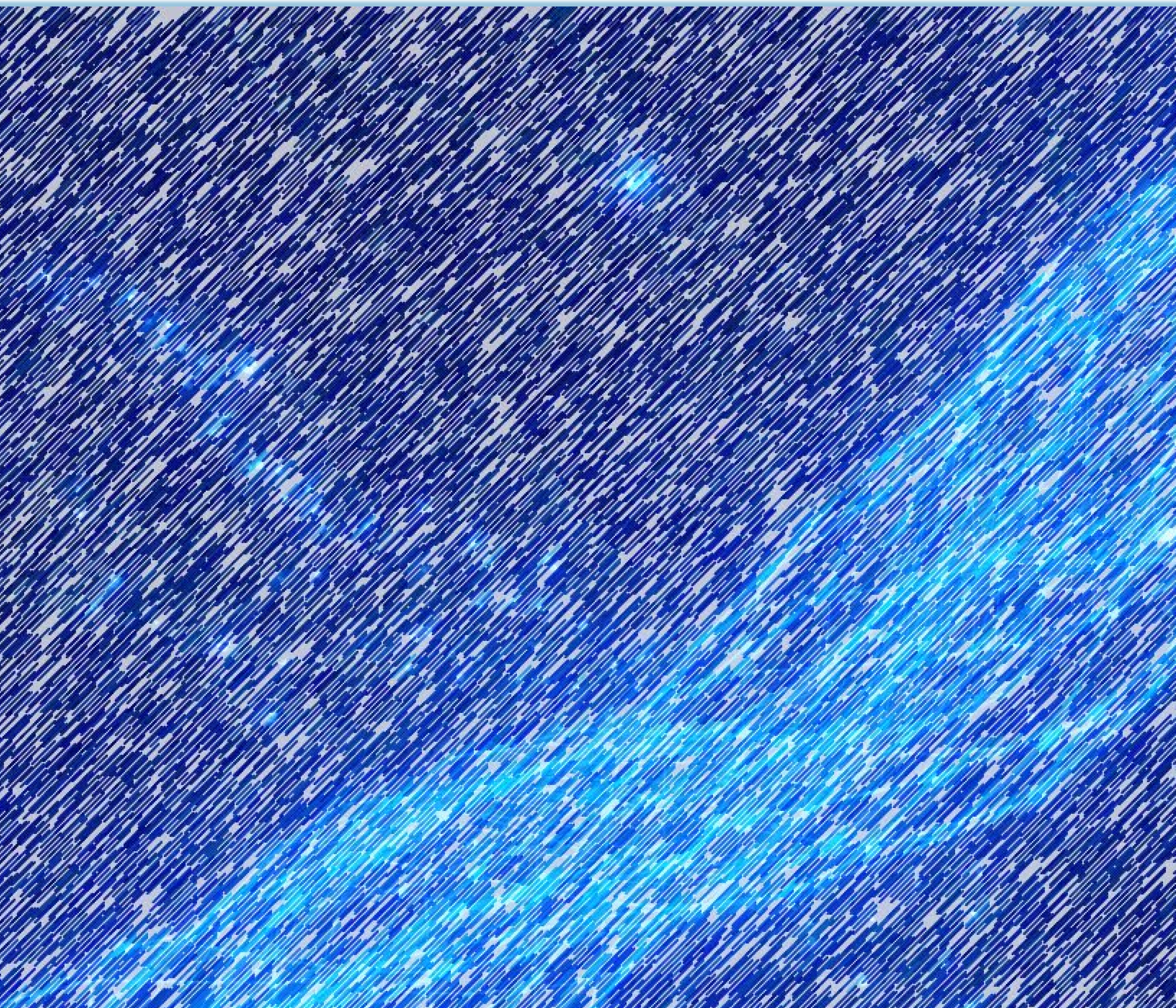

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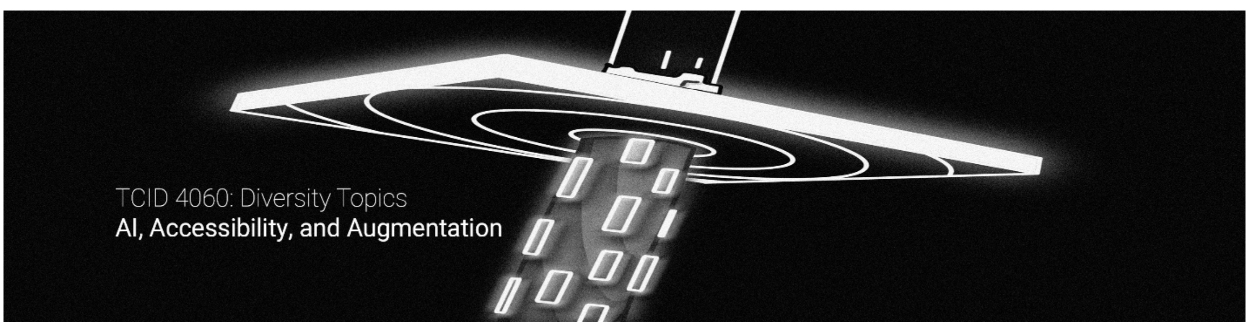
Introduction

In spring 2026, I taught "TCID 4060: Diversity Topics: AI, Accessibility, and Augmentation," a senior-level seminar in Technical Communication and Information Design at the University of Colorado Colorado Springs. I started with a simple enough premise: emerging technologies should not be treated merely as tools to adopt or reject, but as sociotechnical systems that reshape how people communicate, work, move, think, remember, and understand themselves. Across the semester, students examined artificial intelligence, digital accessibility, and augmentation as overlapping areas of professional practice and critical inquiry. They used emerging tools, studied the systems surrounding them, and considered the responsibilities technical communicators and UX practitioners assume when they help design, explain, evaluate, or normalize new technologies. This sequence also extended my ongoing research into AI literacy, which has shown that students develop more durable and professionally useful understandings of emerging technologies when they participate in testing, questioning, and shaping how those technologies are used in the classroom (Strubberg et al., 2024).

The purpose of this paper is to introduce the course and concepts that informed the student projects you will find in this digital exhibit. Additionally, I provide an example of augmentation in action and introduce the student projects.



The course was structured as a hybrid seminar, with weekly in-person meetings supported by synchronous remote participation and asynchronous online work. This format encouraged us to move between discussion and practice: reading theory, testing technologies, analyzing accessibility barriers, prototyping systems, and reflecting on the consequences of design decisions. I asked students to work as researchers, designers, critics, and interpreters throughout the semester. They explored how technologies distribute agency, how accessibility can be treated as a central design principle rather than a final compliance step, and how emerging systems may simultaneously expand human possibilities and introduce new forms of dependence, exclusion, surveillance, or risk.

The projects collected here emerged from the final design module of the course and were developed in partnership with the Kraemer Family Library Digital Exhibits initiative and the Center for Research Frontiers in the Digital Humanities. Students were asked to create a small-scale augmentation—design and actually build a prototype—and then interpret that work for a public audience. Their projects could be physical, digital, hybrid, functional, partially functional, or speculative. I did not care about the quality or fidelity of the prototypes. I cared most about how students evaluated the arguments their prototypes made about access, embodiment, care, agency, and human experience. Taken together, these projects show students using technical communication and UX design to participate critically in imagining and shaping technological futures.

Course Foundations

The course considered a number of guiding questions. What is technology? What does it mean to augment a person? Who gets to decide which capacities need to be extended, corrected, or optimized? And what responsibilities do technical communicators and UX practitioners assume when they help make those systems usable, persuasive, and ordinary?

We began with two readings about information and cognition. In "As We May Think," Vannevar Bush (1945), writing post-World War 2 with a sense of optimism, imagines the memex as an "enlarged intimate supplement" to human memory. His essay offers an early vision of cognitive augmentation through expanded access, association, and recall. Nicholas Carr (2008), writing amid the wars in Iraq and Afghanistan and the 2008 recession, complicates that optimism by arguing that networked technologies may reshape how people concentrate, read, remember, think, and, ultimately, interact socially with each other. We used Bush and Carr to introduce Bernard Stiegler's concept of the *pharmakon*. For Stiegler (2013), technology and tools are ambivalent: technology may function as both remedy and poison. A system that expands access to knowledge may also weaken sustained attention. A tool that extends memory may encourage us to externalize it. This pharmacological tension became one of our primary ways of approaching generative AI. AI systems can support brainstorming, translation, design, coding, and access, but they can also produce false information, obscure authorship, flatten linguistic difference, and encourage dependence. I asked students to resist treating AI as either salvation or catastrophe and instead examine what it augments, what it diminishes, for whom, and under what conditions.

Heidegger's "The Question Concerning Technology" deepened that discussion. Technology, he argues, is not neutral, and offers a way of revealing the world (Heidegger, 1977). Through *enframing*, people, environments, and materials become *standing-reserve*: resources made available for use. These concepts helped us examine AI both digitally and environmentally: human writing becomes training data; users become sources of behavioral information; and labor, electrical grids, water, and material infrastructure become inputs to computational systems. So, we asked what AI can produce, and what people, knowledge, labor, and environments it demands us to make available and change.

From there, we considered augmentation more broadly. Duin and Pedersen (2023) describe augmentation technologies as systems that add cognitive, physical, sensory, or emotional capacities to the body or its surrounding environment. This definition allowed students to move beyond implants and science-fiction cyborgs. An augmentation might instead be a memory aid, a redesigned interface, an adaptive workflow, or a small physical object that reduces everyday strain. Digital accessibility gave us a practical way to evaluate those systems. We treated accessibility as an ongoing design practice and human-centered concern instead of a compliance step. Students examined medical, social, and human-rights models of disability and considered how each model changes what designers identify as the problem. Lewthwaite and Swan (2014) further showed us that standards remain culturally and materially situated; compliance cannot replace judgment, participation, or attention to lived experience.

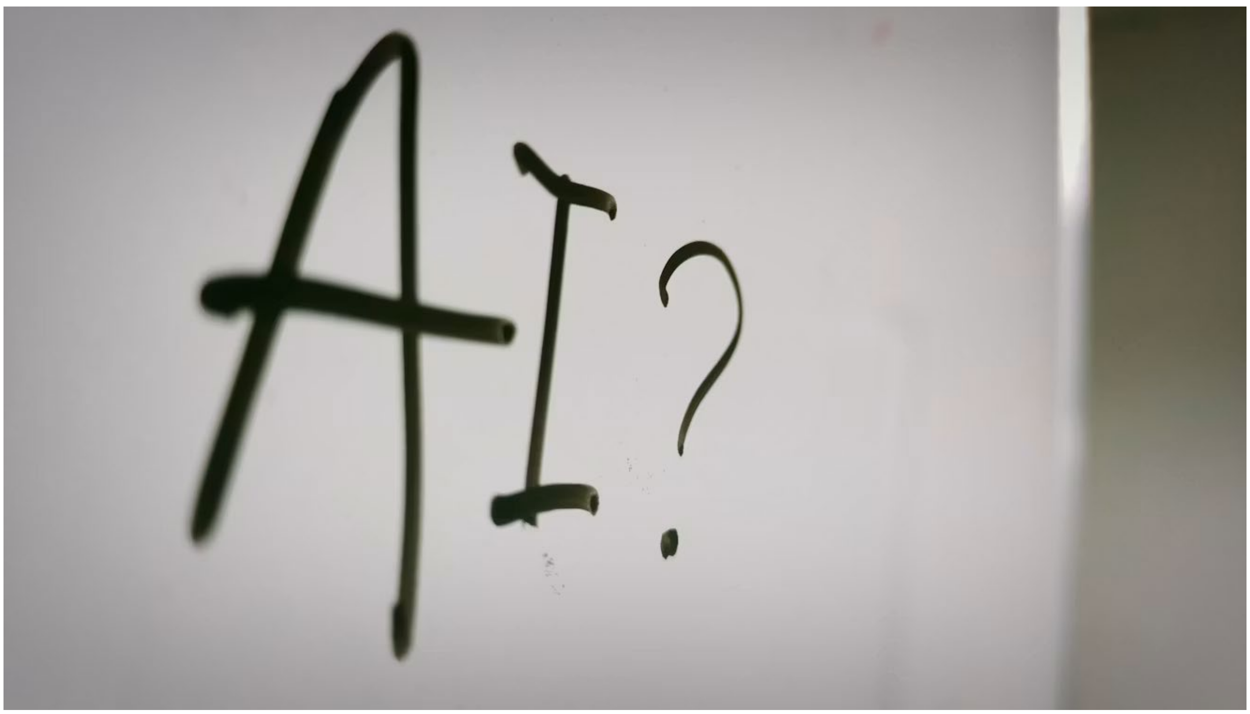
Haraway's cyborg eventually brought these threads together. The cyborg is not simply a human improved by a machine. It represents the unstable boundaries between human and technological, natural and artificial, physical and informational (Haraway, 1987). Students used that instability to question how augmentation technologies shape identity, agency, access, responsibility, and everyday life.

These are just a smattering of the varied readings that informed the course projects.

Course Sequence

I designed the course so that each major project approached augmentation from a different angle and built on the work before it. The sequence moved from designing AI systems, to evaluating digital accessibility, to making augmentation prototypes, and finally to inhabiting speculative futures shaped by these technologies.

The first major project focused on AI-augmented workflows. Students identified a problem that might be addressed through a custom chatbot, designed the bot's instructions, iterated its behavior, supplied relevant documentation, and tested its responses. They also wrote explainable AI statements describing what the bot could do, what information it relied on, where its limitations were, and how users should interpret its outputs. The project required students to see AI as a designed system shaped by instructions, source materials, and human judgment. This approach followed work I have done elsewhere in the UX classroom, where students examined AI as both a tool and an object of research, connecting the user-facing experience to the instructions, information architecture, and technical systems operating behind it (Strubberg & Blackburn, 2025). It also returned to the pharmakon: AI could improve access and efficiency while introducing errors, uncertainty, and new labor.



The second project focused on digital accessibility. Students audited digital materials and wrote professional reports explaining the barriers they found and the changes they recommended. During the unit, we also worked with the Faculty Resource Center and examined UDOIT, an AI-assisted accessibility scanning tool used within Canvas. UDOIT helped students see how automated systems can identify recurring problems quickly, but it also showed the limits of automated evaluation. A scan could flag missing alternative text, heading errors, or contrast concerns, but it could not fully judge whether information was understandable, navigation was intuitive, or users felt supported. The work required human intervention in collaboration with the digital tools.

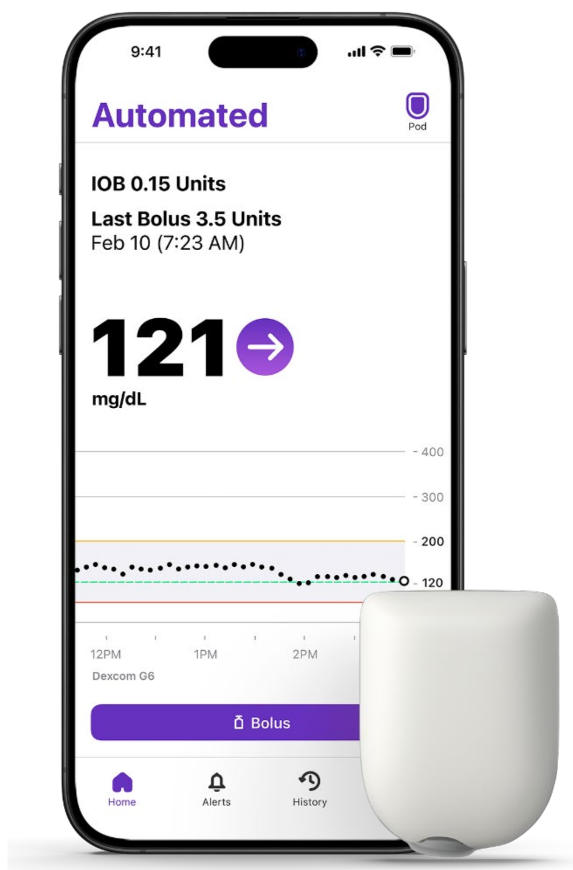
The final design project brought these strands together through the Augmentation Prototype and Digital Exhibit project. Students identified a human need, friction, or constraint and had to make a prototype. It could be a paper sketch, wireframe, physical model, digital interface, coded tool, 3D model, or high-fidelity design. I did not care whether it worked perfectly or at all. I cared that students engaged in making and used that process to discover the affordances and constraints of their ideas in relation to TCID and the course themes. The projects generally became operational prototypes, practical designs that could be built with current technologies, or speculative augmentations. Students then interpreted their work for a public audience through the final deliverables you find in this collection in a variety of ways.

The course concluded with students creating personas of their future selves in an augmentation-saturated world and inhabiting those personas during a three-hour tabletop one-shot that I designed alongside AI. Through this role-playing experience—fueled by junk food, desserts, and refreshments—the various course themes and projects became an embodied environment students had to navigate.

A Kairology of Augmentation for Type 1 Diabetes

My own life with type 1 diabetes gave the course themes a personal and practical center, and I offer some thoughts on diabetes treatment technologies here as a concise example of augmentation in action. Diabetes is often described through numbers: blood glucose readings, carbohydrate ratios, insulin doses, time-in-range percentages, and A1C results. These metrics matter, but type 1 diabetes cannot be understood through numbers alone. It is also an embodied system of technologies, routines, decisions, environments, institutions, and relationships. Georges Canguilhem (1991) argues that disease is not simply a lesser version of health but “a new dimension of life” (p. 186). A person with a chronic disease must establish new norms within a changed and sometimes narrowed environment (p. 185). This idea has shaped how I understand diabetes. Treatment does not return me to an untouched state of normalcy. I instead continually build a workable relationship between my body and the world around it. Meals, exercise, sleep, work, travel, stress, illness, insurance, and access to supplies all become part of the treatment system.

The history of diabetes also demonstrates that augmentation technologies can and often do transform what a disease is and what it means to live with it rather than merely improving it. Before insulin, type 1 diabetes was rapidly fatal. Insulin made survival possible, but it also, as Feudtner (2003) explains, “transmuted diabetes into an evolving biological entity” and radically altered the lives of patients, families, and healthcare workers (p. 25). The remedy created new conditions of life and new forms of labor, risk, and dependence. Later technologies continued this transmutation. Blood glucose meters allowed patients to produce clinical data at home, insulin pumps moved treatment toward continuous delivery, continuous glucose monitors transformed glucose into a stream of data. Each advancement redistributed agency across a larger assemblage of patient, sensor, pump, algorithm, phone, clinician, manufacturer, insurer, and family. Duin and Pedersen (2023) describe agency in augmentation systems as distributed across human and nonhuman participants rather than belonging to the user alone (pp. 78–80). Diabetes technology makes that distribution especially visible: the system may act for me, alert me, advise me, or change insulin delivery, but I remain responsible for interpreting and responding when it fails. This responsibility makes apparent a contradiction in the rhetoric of diabetes management. Stone (1997) argues that managed care attempted to grant patients agency, though still within the constraints of biomedical discourse (pp. 214–215). Bennett (2009) is more critical, arguing that management rhetoric can project blame onto the diabetic body and leave the individual responsible when treatment fails (p. 3). Augmentation can therefore create agency while also intensifying expectations for self-surveillance, discipline, and compliance.



Duin and Pedersen's (2023) discussion of socio-ethical consequences further shows why these systems cannot be evaluated by clinical effectiveness alone. Embodied augmentation technologies collect deeply personal data and raise questions about privacy, consent, control, safety, fairness, and professional responsibility (pp. 124–128). For a person with diabetes, the body becomes more and more readable but also increasingly tracked, transmitted, evaluated, and made available to institutions and corporations.

Technological shifts in care may be best understood through what Segal (2005) calls kairology: "the principle of contingency and fitness-to-situation" (p. 22). Medical and rhetorical shifts occur as responses to changing situations (p. 17). A treatment is neither advanced nor outdated; rather, value depends on the person, moment, environment, infrastructure, and support surrounding it. A pump may provide freedom in one setting and become a visible burden in another. For example, an alarm may prevent an emergency while interrupting sleep, work, school, or extracurricular activities. More data may support better decisions while producing anxiety and continuous demands for attention.

For work in technical communication and information design, diabetes provides a clear example that augmentation not only a device. It involves interconnected and ever-evolving relationships between interface design, alarm language, instructions, accessibility, data ethics, maintenance, and the systems determining who can obtain and sustain the technology. The central concern becomes whether a particular technology fits a particular life, at a particular moment, under particular conditions.

Augmentation Made Visible

Student projects developed along three broad paths: operational prototypes that could already be used, practical designs that could be built with current technologies, and speculative augmentations that imagined future possibilities. These categories often overlapped. Beyond classification, each project made a particular human need visible and tested what augmentation and accessibility technologies might offer in response.

Many projects sought solutions for everyday physical frictions. Caleb Balduff's "Blister Blaster" reimagined medicinal blister packaging for people with arthritis, chronic pain, tremors, or limited mobility. His low-fidelity lever press showed how a simple mechanical intervention could relocate effort from the user's hands into the design. Joslyn Bayley's "Between Uses" offered a modular glasses chain and magnetic frame charms for people who frequently remove or misplace their glasses. Chris Rexroad's "Fidget Crimper" examined wearable pressure as an unobtrusive aid for ADHD focus, while Torian Stiles' "Wrist Stability Guard" used AI-assisted 3D modeling to develop support for reducing pain during computer work. These projects framed augmentation as simple solutions for ordinary strain rather than superhuman enhancement.

Other students focused on cognition, memory, and executive function. Mariah Lopez's "Recall & Reflect" became a working mobile-first application for recording memory lapses and identifying patterns, followed by reflection. Sam Garibay's "FocusFlow" combined task management, energy check-ins, timers, brain dumps, analytics, and AI-supported task compartmentalization into an executive-function system. Both projects explored how interfaces can shoulder cognitive labor without pretending to eliminate the difficulties they address.

Creative and educational augmentation appeared in Colin Busak's "Serum Synthesis Guide" and Joshua Bohumil's "VisionCraft 3D." Colin developed an AI-powered sound-design assistant that generated synthesizer settings while explaining how they shaped sound, positioning AI as both a teacher and collaborator. Joshua built a browser-based voxel sculpting tool for users with incomplete achromatopsia, connecting accessible design, creative practice, and AI-assisted coding.

Jessica Chappelle's "Aqualert" and Tatiana Choate's "NIMBI" moved into systems of care and monitoring. "Aqualert" combined a child-worn bracelet with a digital application for caregivers intended to detect water exposure, movement, location, and distress. Jessica's physical model and functional interface prototype demonstrated the system's practical appeal while raising questions about reliability and continuous monitoring. "NIMBI," a speculative companion drone, imagined ambient care through biometric sensing, reminders, alerts, and routine support while also confronting surveillance, dependency, privacy, and sustainability. Julie McMullen's "Aether" considered how to restore proprioception and kinesthetic awareness through speculative design. The project extends the collection's concern with bodies that do not always receive or process sensory information in expected ways and asks how technology might support movement, perception, and bodily trust.

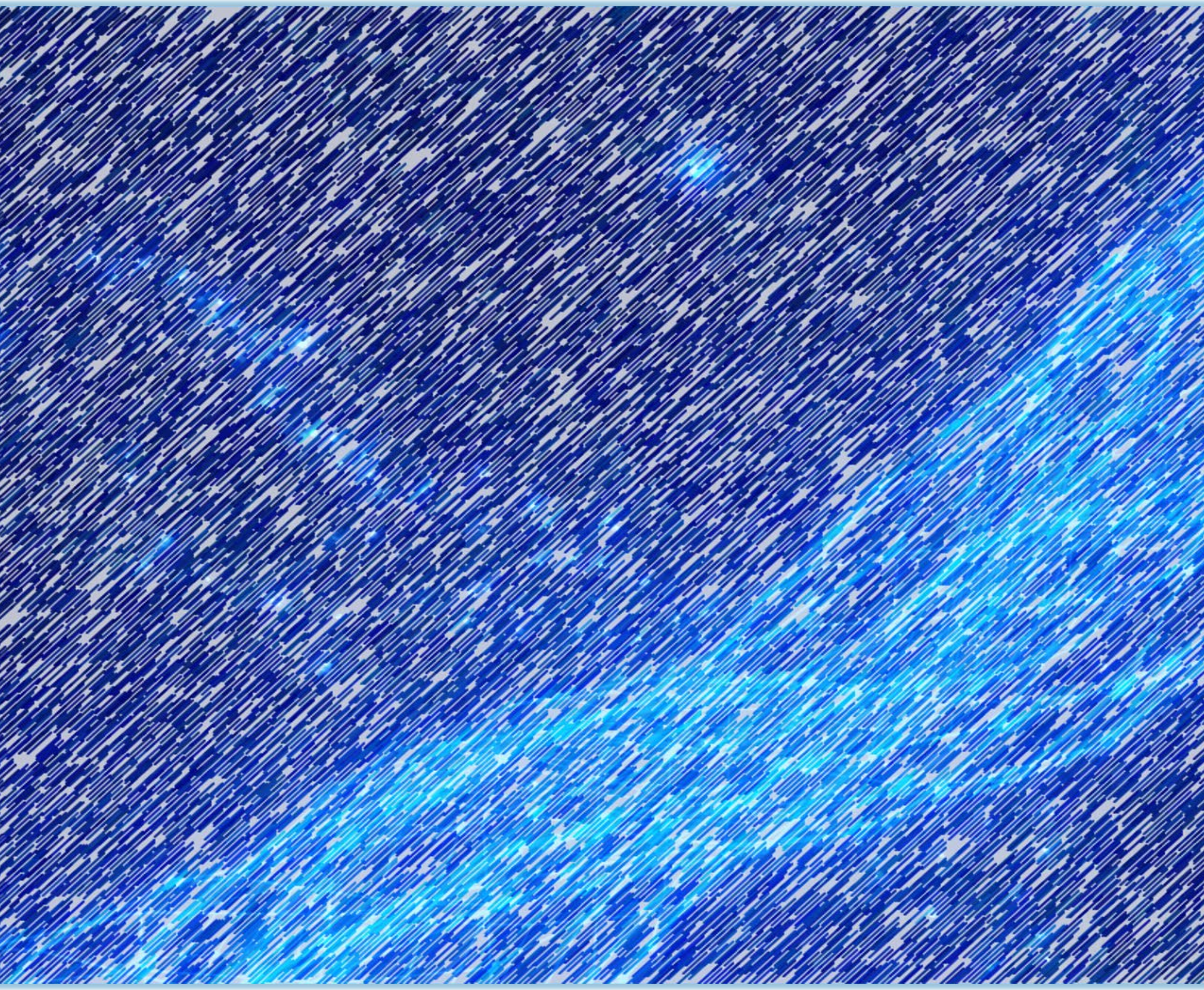
Across the collection, students resisted the idea that augmentation always makes a person "better." These projects instead looked at people in context, asking what might be needed in a particular context and what new affordances, constraints, labor, and risks a designed response would introduce. The prototypes make those questions tangible, presenting augmentation as a series of choices about bodies, attention, care, access, creativity, and everyday life.

Conclusion

The projects in this collection consider augmentation as a series of situated design choices. Each prototype responds to a particular need, body, environment, or friction. Each design also carries its own constraints, assumptions, and risks.

Throughout the semester, students worked across AI, accessibility, UX research, critical theory, and making. They designed systems, audited interfaces, built prototypes, tested ideas, and reflected on the social consequences of their choices. In doing so, they learned that technical communication and information design are not secondary to technological development. These fields and methods help determine how systems are explained, trusted, accessed, maintained, interpreted, and adopted by people. They also help determine whose needs are recognized and whose remain invisible.

This collection reflects that responsibility. Some projects are modest and immediately practical. Others imagine systems that may never be built in the forms presented here. All of them ask readers and viewers to look closely at the relationships between people and technologies and to consider what happens when a design moves from concept into everyday life. The guiding question of the final project was, “How might augmentation technologies reimagine accessibility and user experience?” The student work collected here does not offer a single answer. Instead, it offers a set of possibilities, cautions, and provocations. It invites us to consider not only what technologies can do, but what kinds of lives, relationships, and futures we want them to help create.



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